Lab Report 2: The PIFA Antenna

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1. Introduction

The previous lab served as our entry point to the world of antennas and the fundamental principles of antenna theory. From the roots, we studied how an antenna works, what electromagnetic principles allow it to emit radio waves and how the radiated power is characterised. At this point, we are well aware of how relevant impedance matching is to have a well-adapted antenna with a low reflection coefficient, or how useless it can become having a well-adapted antenna when its Gain, F/R and F/B ratios are not well optimised to produce a good radiation pattern, aligned with the characteristics that our practical application requires.

Nonetheless, up to this point, we have focused primarily on wire antennas, which is natural, as they are the most fundamental type of antenna. They are useful to begin learning with because the electrical principles can be related to them very easily. However, wire antennas come with several limitations that can impact their usability in certain applications.

One major drawback is their size and shape constraints. Particularly at lower frequencies, wire antennas require significant physical length to remain resonant, making them impractical for modern compact devices. For instance, recall that the original size of the dipole in our Yagi Uda antenna was set to $LA=0.5\lambda=0.5\cdot3\cdot10^8/2.4\cdot10^9=0.0625m$. However, if one wanted to radiate at a much lower frequency, say 4MHz, then $LA=0.5\lambda=0.5\cdot3\cdot10^8/4\cdot10^6=37.5m$ which is completely impossible to achieve in real life. Additionally, modern devices such as smartphones not only demand small antennas but also require specific form factors to enhance the overall aesthetic appeal of the device.

It is to address these limitations that the so-called patch antennas, such as the Planar Inverted-F Antenna (PIFA), have gained significant popularity in recent years, particularly in compact communication systems. Patch antennas offer smaller form factors, making them ideal for devices like smartphones and WiFi routers. Additionally, patch antennas have much more customizable radiation patterns, allowing for directed or omnidirectional radiation depending on the requirements.

For that reason, in this lab, we shift our focus to PIFA antennas. We will explore their design, compare them to wire antennas and analyse their performance using the same methodology as before: design and simulation, fabrication, measurements and datasheet.

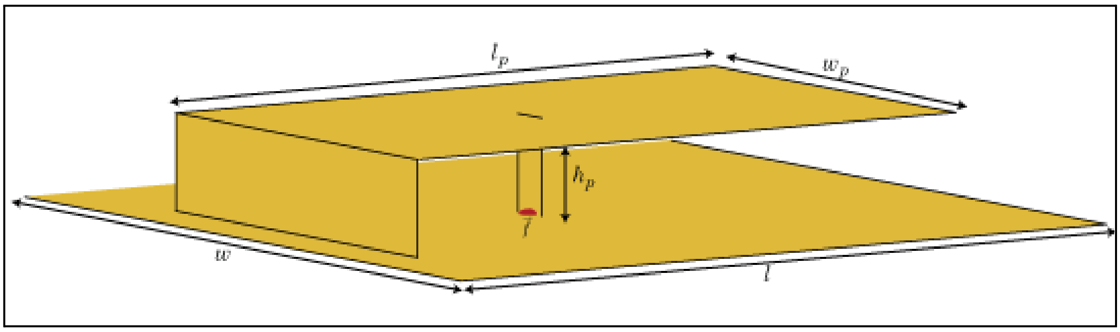


Figure 1: PIFA antenna diagram.

2. Design and simulation of the antenna

2.1. Principles of the antenna design

The Planar Inverted-F Antenna (PIFA) receives its name from the inverted F shape it has when looked at from a vertical plane perspective. Simply put into words, it consists of a monopole antenna running parallel to a ground plane and grounded at one end. The antenna is fed from an intermediate point a distance from the grounded end.

This antenna is notorious for its small form factor that allows it to be printed directly onto circuits using the microstrip format and its omnidirectional radiation pattern. The latter refers to its ability to radiate power equally in all directions perpendicular to an axis. Specifically in the case of our antenna, when drawn in three dimensions this radiation pattern would look like the upper half of a burger bun placed right on top of the ground plane, but we will dive deeper into this later on. This is one of the main differences with the Yagi Uda antenna, which was much more directive, however, in the context of smart devices, omnidirectional radiation makes more sense.

The following diagram shows the main elements of a PIFA antenna:

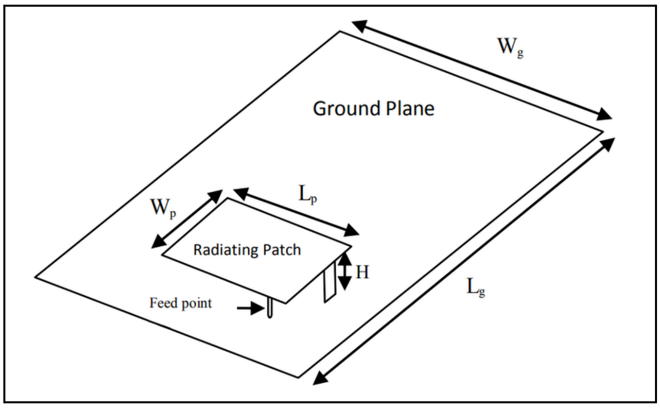


Figure 2: PIFA antenna main elements.

These can be identified as 4 main elements:

- Radiating patch: The main conductive element that emits and receives electromagnetic waves (upper sheet of metal).
- Ground plane: A conducting surface that enhances radiation efficiency and impedance matching (lower sheet of metal).
- Shorting plate: The element connecting the unit to the ground plane, lowering the resonant frequency and reducing the antenna's size.
- Feed point: The element supplying the signal to the patch, typically placed at an optimised position to achieve 50Ω impedance matching.

Moreover, thanks to some intricate mathematical antenna properties, when the width of the shorting plate equals W_P, the resonance frequency can be set by

$L_P=\lambda/4$. Also, note that following the advice of the professor, we will place the feed point in the middle point of the W_P-length side that does not contain the shorting plate. Lastly, the PIFA in this lab is designed to operate at 2.4 GHz, a common frequency for WiFi, with an impedance of 50Ω to ensure proper impedance matching and minimize reflections. This will allow us to clearly see and highlight the differences with a wire antenna such as a Yagi Uda.

2.2. Simulation using 4NEC2

We now reintroduce the concept of simulation, where we use a program named 4NEC2 to be able to simulate the real-life behaviour of an antenna with a specific set of characteristics. As we saw in the previous lab, this is useful as it allows us to iterate over different versions of the parameters very fast, without needing to build and measure the antenna in real life and gives us access to tools such as the optimiser. Ideally, the simulation resembles very closely the real-life conditions, but as we already saw in the last lab this is rarely the case.

On this occasion, the professor included with the handout of the laboratory a .NEC file containing an initial setup for our PIFA antenna. We present now all the variables in this file and clarify some necessary points:

Symbols

Nr	Symbols and equations
1	DW=3.125e-3
2	DL=3.125e-3
3	NL=10
4	NW=20
5	NS=20
6	L=DL*NL
7	W=DW*NW
8	H=6.25e-3
9	F=0
10	R=0.0005

Figure 3: PIFA antenna initial NEC file.

The different variables represent the following:

- $W=NW\cdot DW=20\cdot 3.125e\text{-}3=62.5e\text{-}3$m. (width of the radiating patch)
  - DW is the width of a segment in the antenna structure.
  - NW is the number of segments in the width direction.
- $L=NL\cdot DL=10\cdot 3.125e\text{-}3=31.25e\text{-}3$m. (length of the radiating patch)
  - DL is the length of a segment in the antenna structure.
  - NL is the number of segments in the length direction.
  - Notice that $\lambda=3\cdot 10^8/2.4\cdot 10^9=0.125$m, so by default $L=\lambda/4$.
- H=6.25e-3m.
  - H is the height of the radiating patch from the ground plane.
- F = 0m
  - Feed position along the x-axis.
- NS=20

    - Number of total segments. This parameter is added for construction reasons but will not be relevant to our study. Following the professor's advice, this parameter should always be equal to NW, for correct simulation behaviour.
- R=5e-4
    - The radius of the wire used to simulate the mesh representing the radiating patch in 4NEC2. This parameter is added for construction reasons but will not be relevant to our study, we assume the mesh works just as a sheet of metal.

This setup yields the following PIFA:

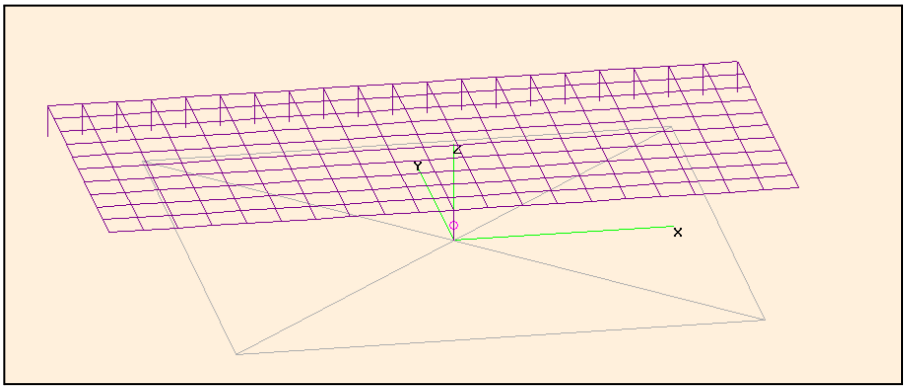


Figure 4: PIFA antenna initial distribution.

Notice how the feed point is set at (0,0). Then, the radiating patch is defined on the x-axis from -W/2 to W/2 and on the y-axis from 0 to L. Also, realise that L=λ/4, however, this antenna is not currently completely correctly set up to resonate at 2.4GHz. Figures 5 and 6 below are proof of this:

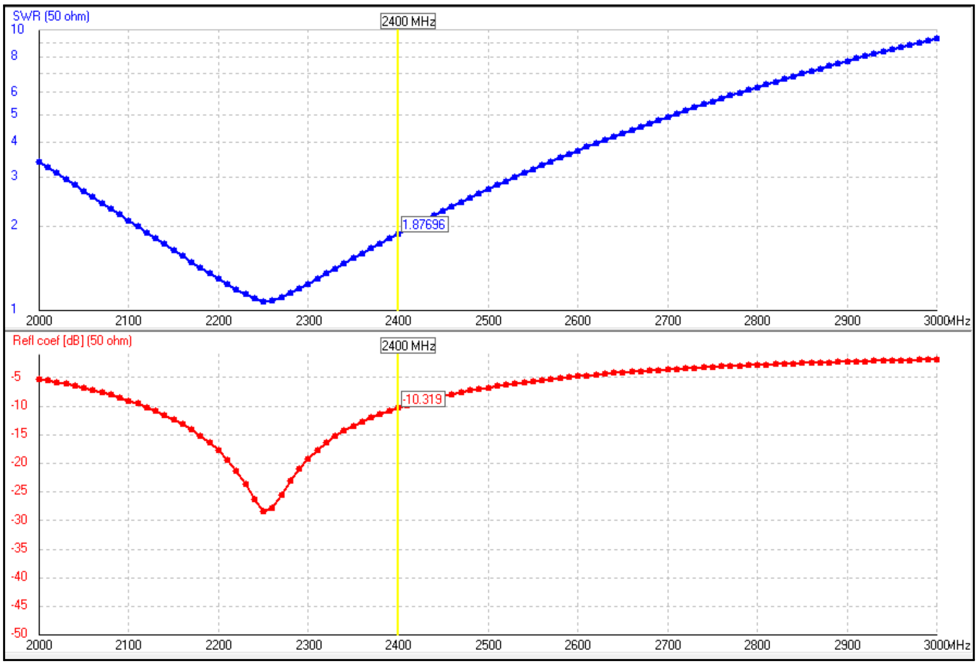


Figure 5: PIFA antenna with initial parameters reflection coefficient graph.

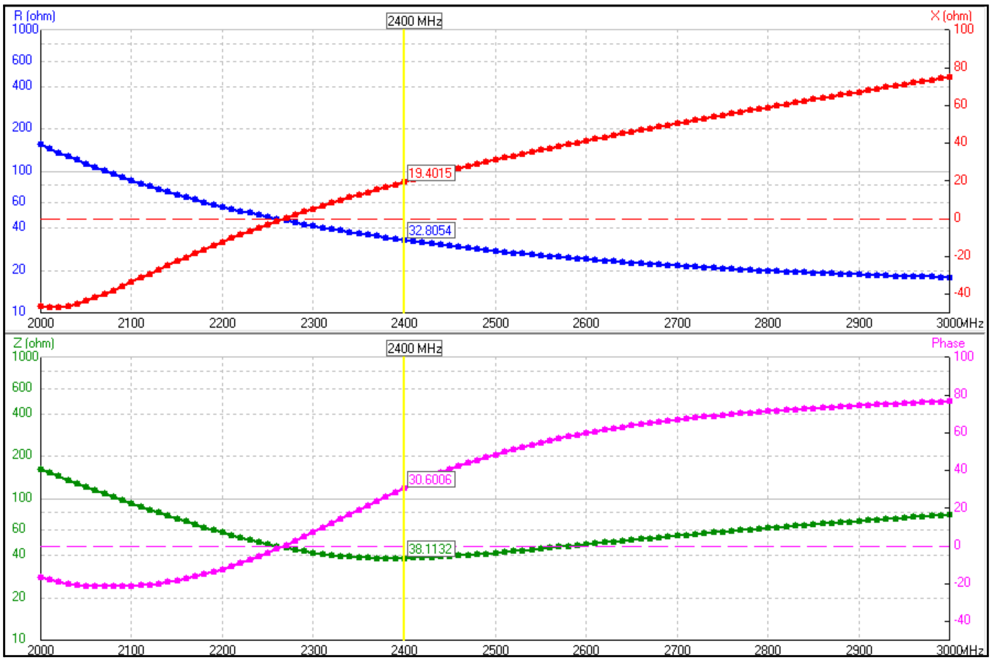


Figure 6: PIFA antenna with initial parameters impedance graph.

Figures 5 and 6 clearly show that this antenna is not yet well adapted to work in the 2.4GHz frequency range, mainly because of the need for frequency scaling and the fact that $L=\lambda/4$ is a default value which usually guarantees a decent performance from the antenna but which might need optimisation. More on this later.

As a side note, consider that even though in real life (see Figure 2) the ground plane has to be of finite size, in theoretical domains, and therefore in our simulation, the ground is considered infinite, which is why only the dimensions of the radiating patch are specified, see the Figure 7 below:

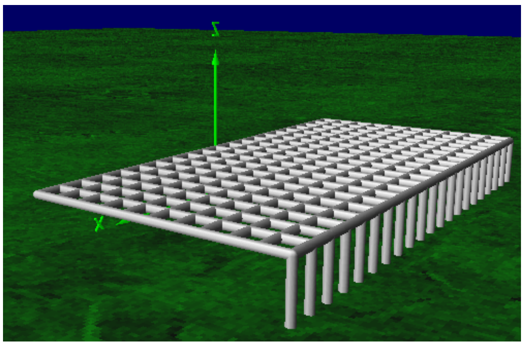


Figure 7: PIFA antenna with initial parameters 3D graph.

Now we move onto the stage of optimisation, where we will adapt the antenna to work in the desired frequency range and carry out other improvements.

2.3. Optimisation and frequency scaling

At this point, we have correctly explained and understood the parameters and main properties of our PIFA antenna, and now it is time to improve the antenna behaviour by carrying out optimisation and frequency scaling. However, the same question as in lab 1 arises, and that is, what are we optimising? It turns out that the answer to this question is slightly different to the answer of the previous laboratory. We present the radiation diagrams of the current antenna as they will be useful in answering this query.

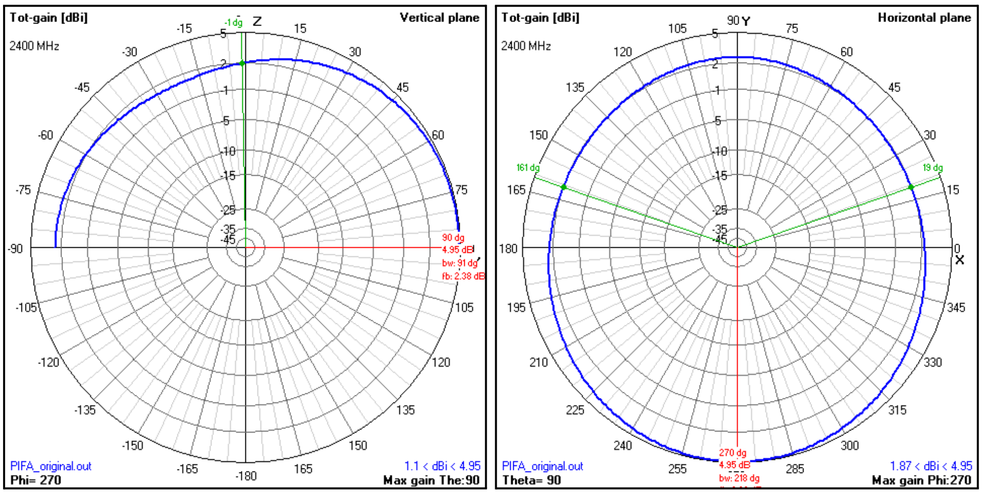


Figure 8: 2D Radiation diagrams - PIFA with default values.

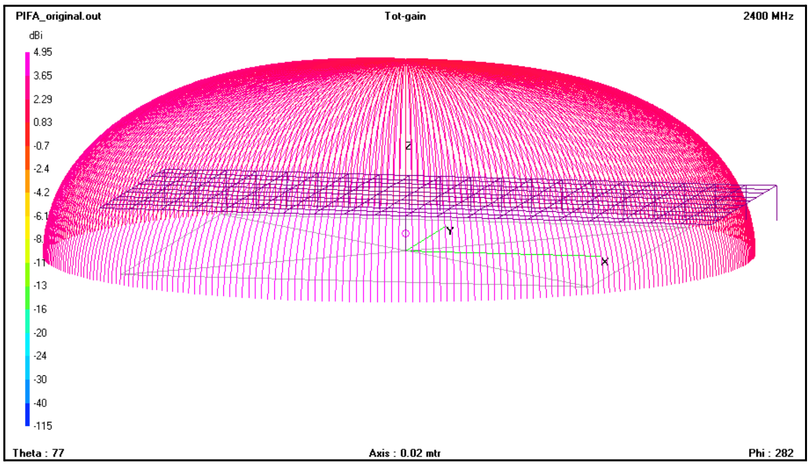


Figure 9: 3D Radiation diagram - PIFA with default values.

For now, we will ignore the fact that this antenna is not fully adapted to 2.4GHz because its diagrams already present a very interesting property: Unlike the Yagi Uda, which focused the optimisation efforts on retaining a good reflection coefficient while improving the Gain and radiation pattern, in this case, we can see how the PIFA antenna already has a hard-to-improve radiation diagram. It radiates almost perfectly in the horizontal and vertical planes with omnidirectional properties even though the antenna is not scaled to resonate on the correct frequency (the radiation patterns above are set up for 2.4GHz). Also, the gain presented is already very good for these types of antennas so it can hardly be improved.

This is crucial to realise, as the goal in this lab is not so much to improve the gain and radiation pattern, but rather to be able to reduce the size of the antenna as much as possible while retaining the good reflection coefficient and radiation characteristics it already presents.

This makes sense in a context where the antenna does not need to be directive. Rather the opposite, as it would be quite uncomfortable to point our smartphone right towards the nearest cellphone coverage tower to, for instance, call someone. Therefore, in this case, we much rather have a non-directive antenna, which is as small as possible to fit in any device.

The procedure we will follow to do this is very simple and consists of 3 steps:

1. Manual Cut: We will carry out a manual cut on the dimensions of the PIFA antenna (NW, NS) to significantly reduce its size.
2. Frequency Scaling: We will scale the frequency of the antenna to correctly be adapted at 2.4GHz.
3. Optimisation: We will optimise DW, DL and H to fine-tune the performance and be able to retain a good Gain and SWR.

This process is just like cutting a piece of paper to a very specific length: first one cuts the excess paper out roughly using manual scissors, then one positions the paper correctly into a precise cutting machine and lastly, the cutting machine performs precise cuts to adapt the paper to its desired size.

Bear in mind that all this is carried out in a context where L should have a size close to $\lambda/4$=31.25e-3m so that the antenna can correctly resonate at 2.4GHz. That does not mean that L must be a fixed value, instead, it means that L must remain close to this theoretical standard to avoid excessive distortion in other metrics of the antenna, which would ultimately lead to worse performance. It is, as always, a matter of striking the correct balance.

To do this, we carried out different approaches and logic to achieve different results which we can compare. These are listed below and can be separated into 3 different iterations of the 3-step process described above.

2.3.1. First iteration

The first iteration was the more conservative approach we carried out. Essentially, we still did not know how the behaviour of the antenna would respond to drastically reducing its size, so we tried to not go over the edge and reduce NW to 14 and NS to 12.

It is a good moment to point out some things that will carry on later during the following steps and iteration. We will not modify NL. The reason for this is that altering NL would affect substantially the dimensions of L, which is already set up to be $\lambda/4$. Consequently, we leave the task of optimising L to the optimiser, which can fine-tune DL in such a way that the changes are small enough to have a positive effect on performance without completely messing up the adaptation of our antenna.

On the other hand, notice that NW is not equal to NS. This a mistake we came across in our first iteration. What this means in practice is that some excess wires will be floating in space somewhere close to the radiating patch. However, after consulting the professor, these wires should not be having any significant effect on the performance and, in any case, this attempt served as a learning experience for later attempts.

Therefore, after the significant reduction to NW and NS, the antenna symbols look as follows:

Symbols

Nr	Symbols and equations
1	DW=3.125e-3
2	DL=3.125e-3
3	NL=10
4	NW=14
5	NS=12
6	L=DL*NL
7	W=DW*NW
8	H=6.25e-3
9	F=0
10	R=0.0005

Figure 10: PIFA antenna first manual cut NEC file.

The second step of our process is to frequency scale the antenna. Even though the concept of frequency scaling has been introduced and studied in the past laboratory, we simply mention that it is based on scaling the values of all relevant symbols (DW, DL, H) to shift their peak in performance to our desired frequency. NL, NS and NW are not modified because they must be integers and R and F are fixed. We avoid plotting the impedance matching and reflection coefficient of the manual cut antenna to not overload the report with plots. However, we do specify that the antenna was resonating at 2.34GHz, which indicates that the variables must be scaled by a factor of 2.34/2.4=0.975.

Symbols

Nr	Symbols and equations
1	DW=3.047e-3
2	DL=3.047e-3
3	NL=10
4	NW=14
5	NS=12
6	L=DL*NL
7	W=DW*NW
8	H=6.094e-3
9	F=0
10	R=0.0005

Figure 11: PIFA antenna first frequency scaling NEC file.

Then, after the relevant variables have been scaled, it is possible to check the impedance matching and reflection coefficient plots to ensure that our antenna with the manually cut dimensions is well adapted to 2.4GHz:

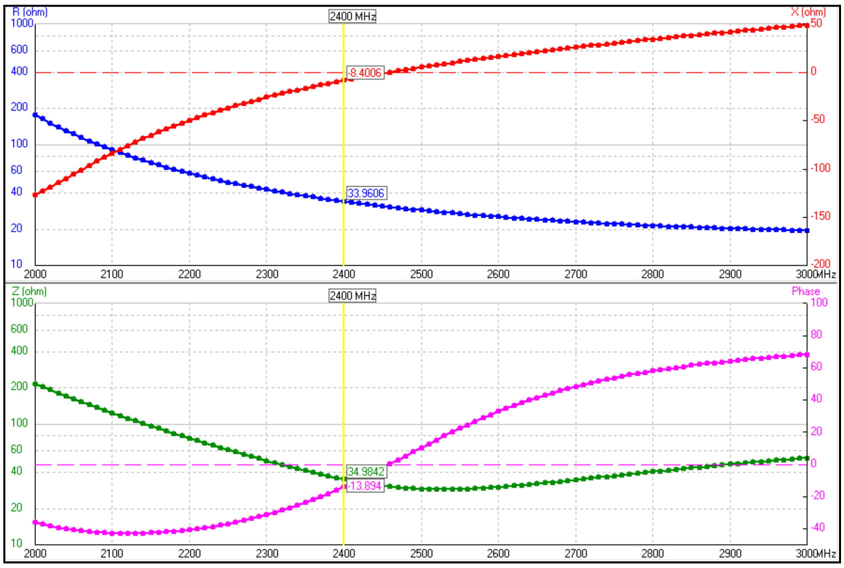


Figure 12: PIFA antenna first frequency scaling impedance graph.

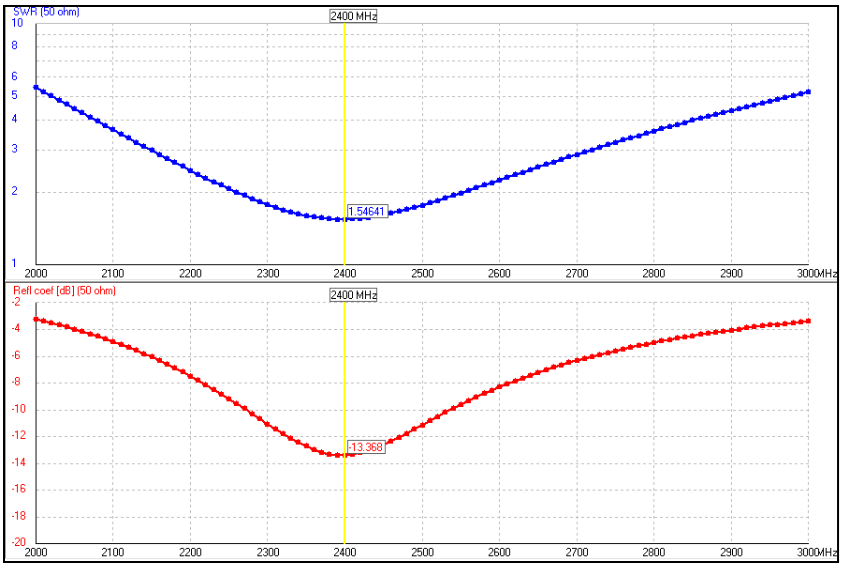


Figure 13: PIFA antenna first frequency scaling reflection coefficient graph.

As we have explained, Figures 12 and 13 showcase a well-adapted antenna at 2.4Ghz, with a decent reflection coefficient of -13.37dB and impedances of 33.95Ω for the real part and -8.4Ω for the complex part.

This performance is by no means perfect and these values can and will be optimised later, but before, we must take a look at the radiation pattern produced by this antenna:

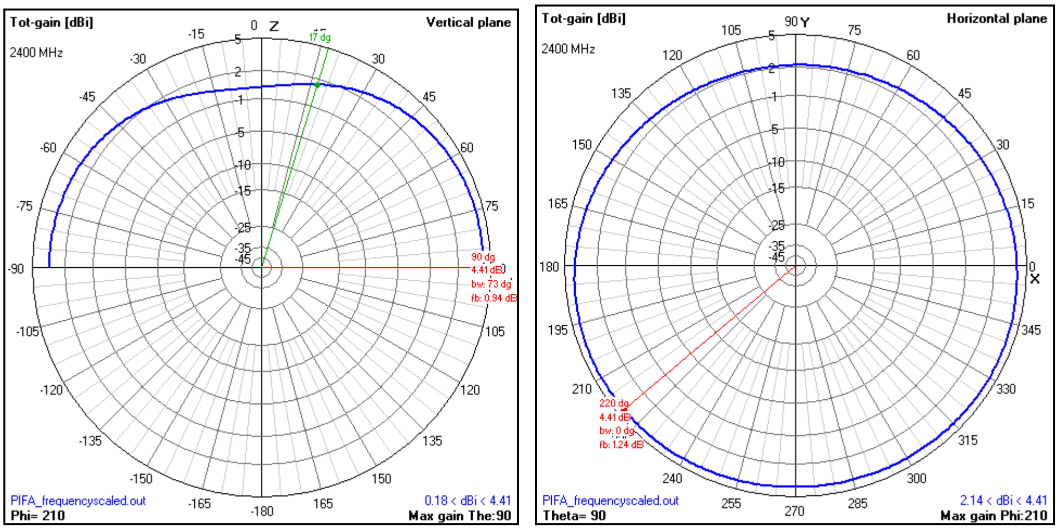


Figure 14: 2D Radiation diagrams - PIFA with first frequency scaling.

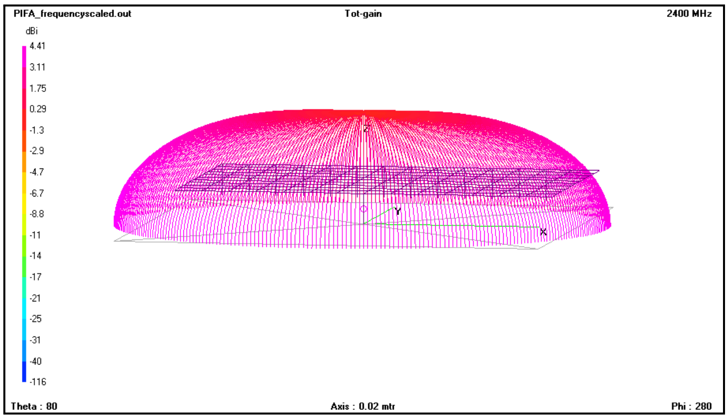


Figure 15: 3D Radiation diagram - PIFA with first frequency scaling.

Figures 14 and 15 show patterns almost identical to those of the original antenna. This was to be expected, as we had already mentioned that the pattern of the non-altered antenna was close to optimal and it was not the main focus of the simulation. In this case, the beamwidth has stayed essentially the same but the direction of maximum radiated power has dropped from 4.95dB to 4.41dB. This is a compromise one would most likely make, as we have already been able to reduce the size of the antenna significantly and have seen little negative effect on performance.

The next step is to carry out optimisation on this frequency-scaled antenna to fine-tune it even further and improve its performance. In this lab, the decision on what to optimise is not that complex, as we have already explained that we aim to retain good Gain and coefficient of reflection (SWR) while reducing significantly the size of the antenna. For that reason, for this and later optimisations, we will optimise Gain and SWR (with the same relevance) for the variables DW, DL and H, which are the ones that can be safely altered by fractions of a millimetre without compromises, as we have explained before. The results yielded are the following:

Symbols

Nr	Symbols and equations
1	DW=3.278e-3
2	DL=2.833e-3
3	NL=10
4	NW=14
5	NS=12
6	L=DL*NL
7	W=DW*NW
8	H=7.022e-3
9	F=0
10	R=0.0005

Figure 16: PIFA antenna first optimisation NEC file.

Comparing the sizes to those of the frequency-scaled antenna, we can see that DW and H have grown while DL has decreased. As always, the changes are millimetric, but as we will see, they play a crucial role in maximising the potential of our antenna. The results are as follows:

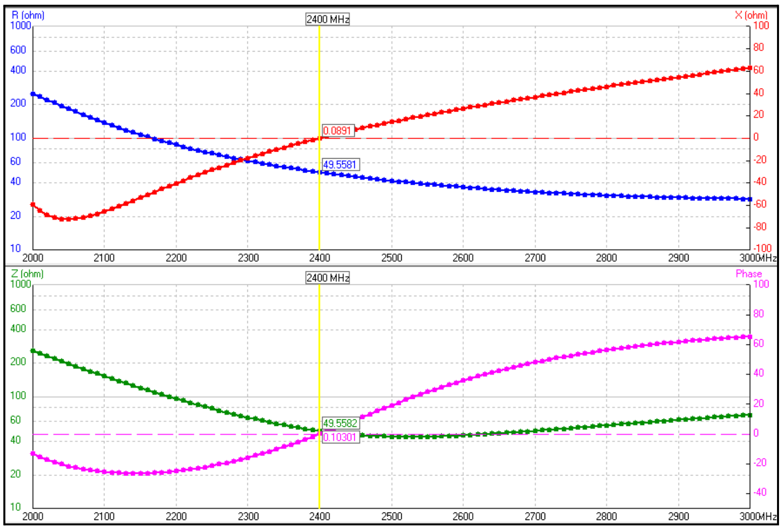


Figure 17: PIFA antenna first optimisation impedance graph.

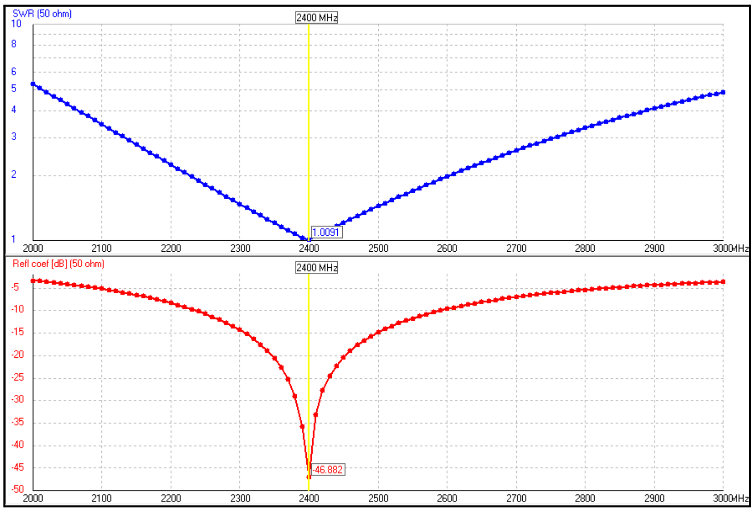


Figure 18: PIFA antenna first optimisation reflection coefficient graph.

Surprisingly, Figures 17 and 18 showcase a very significant improvement over the frequency-scaled version of our antenna in terms of impedance matching and coefficient of reflection. The impedances are very well matched at 2.4GHz with values of 49.56Ω for the real part and 0.1Ω for the complex part. Moreover, the coefficient reflection achieves a value of -46.88dB. These results seemed almost too good to be true since during the previous lab we did not see such clear performance improvements after optimisation. We move on to analyse the radiation pattern and Gain results of the optimisation:

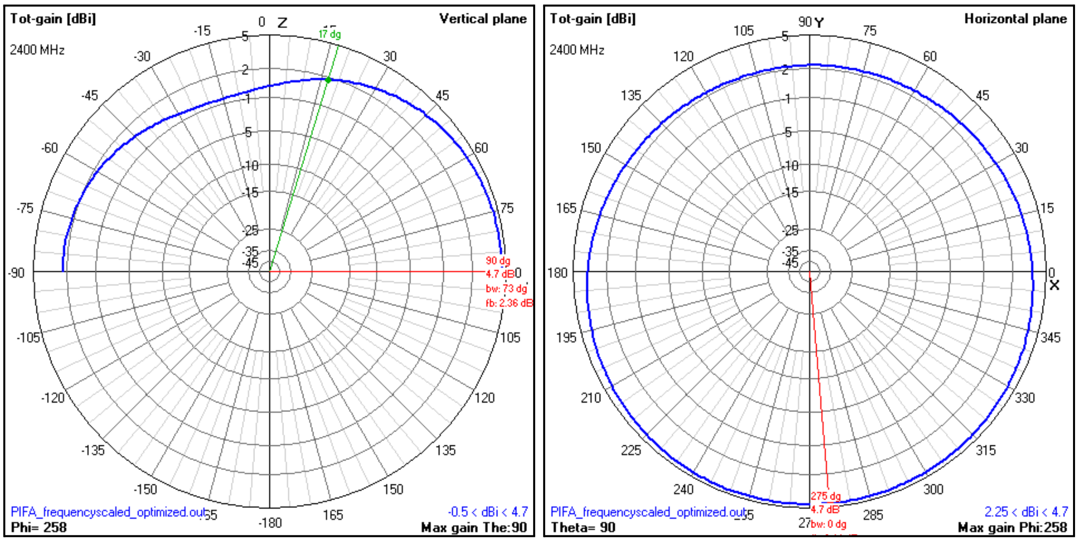


Figure 19: 2D Radiation diagrams - PIFA with first optimisation.

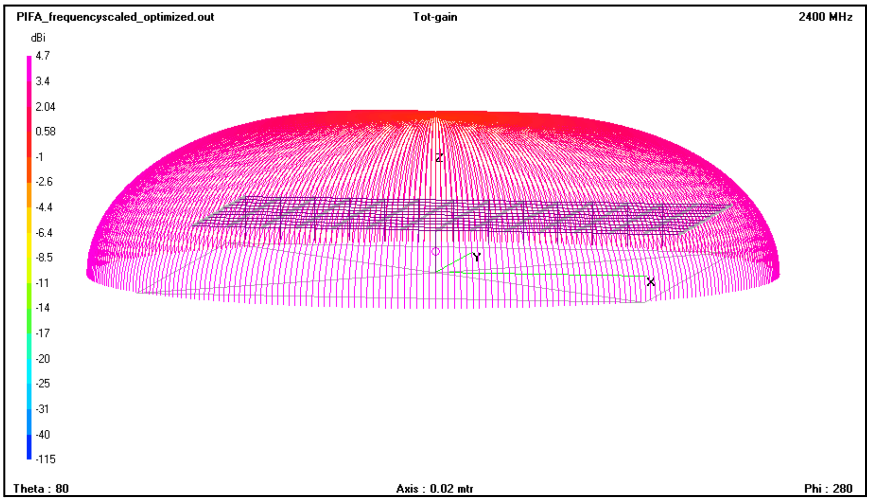


Figure 20: 3D Radiation diagram - PIFA with first optimisation.

Figures 19 and 20 are less surprising than the previous ones, as at this point we were pretty much convinced that we would not see great alterations in the Gain and radiation pattern of our antenna. Nonetheless, we still identify sensible improvements over the

frequency-scaled version, with wider beamwidths in both directions and an increase in maximum power radiated from 4.41dB to 4.7dB
The results of this first iteration were very satisfactory, and the resulting antenna was more than a good performer. In fact, we must admit that these results will hardly translate into reality: the assumption of an infinite ground plane and the precision of the cuts that would be needed to create this antenna are two aspects hard to replicate in real life. Additionally, we commented at the start of the iteration that having a value of NW different to NS could affect performance, and even though in the end it did not look as so, we wanted to ensure that was not the case.

It is for all these reasons that we decided to carry out a second iteration, following the same basic steps but fixing our mistakes and being more ambitious.

2.3.2. Second iteration

As the closure of the first iteration explains, in this second iteration we wanted to correct mistakes and be more ambitious in the size reduction of the antenna, which at the end of the day is the main goal for this laboratory. Note that we started again from the original antenna and not from the previous iteration to avoid carrying any errors.

To do so, we performed a stronger manual cut reducing the values of both, NW and NS to 8. This is 60% smaller than the original version of the antenna, so we were pretty excited to make it work.

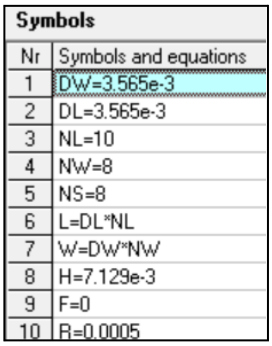
Symbols

Nr	Symbols and equations
1	DW=3.125e-3
2	DL=3.125e-3
3	NL=10
4	NW=8
5	NS=8
6	L=DL*NL
7	W=DW*NW
8	H=6.25e-3
9	F=0
10	R=0.0005

Figure 21: PIFA antenna second manual cut NEC file.

As before, once our antenna had been correctly simulated with the manual cut, we used frequency scaling to adapt its performance to our 2.4GHz requirements. In this case, the antenna was better adapted to 2.76GHz so the scaling factor applied to the variables was 2.76/2.4=1.15.

Nonetheless, after this first frequency scaling we realised that the antenna was not yet perfectly matched, for that reason we decided to carry out frequency scaling again. This time the factor was 2.38/2.4=0.992 and the results this yielded are the following:



Symbols

Nr	Symbols and equations
1	DW=3.565e-3
2	DL=3.565e-3
3	NL=10
4	NW=8
5	NS=8
6	L=DL*NL
7	W=DW*NW
8	H=7.129e-3
9	F=0
10	R=0.0005

Figure 22: PIFA antenna second frequency scaling NEC file.

Having understood from the previous iteration that only DW, DL and H are frequency scaled, the values in Figure 22 are no secret and only represent the result of multiplying the original values by the two frequency scaling factors. The results of this antenna configuration are:

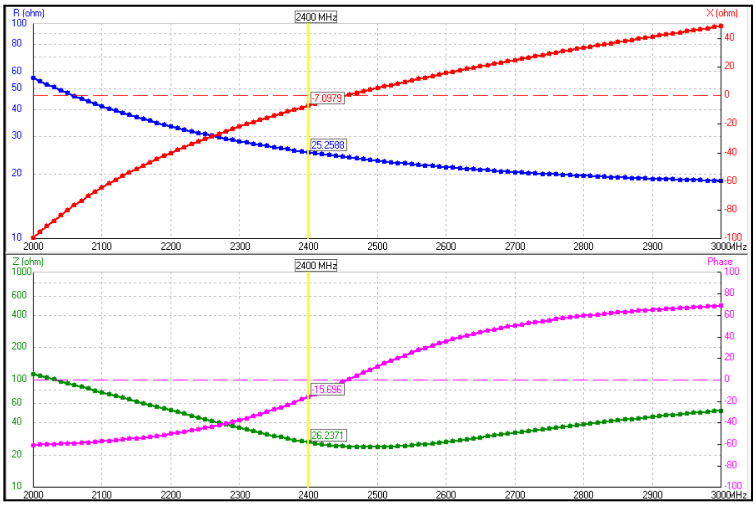


Figure 23: PIFA antenna second frequency scaling impedance graph.

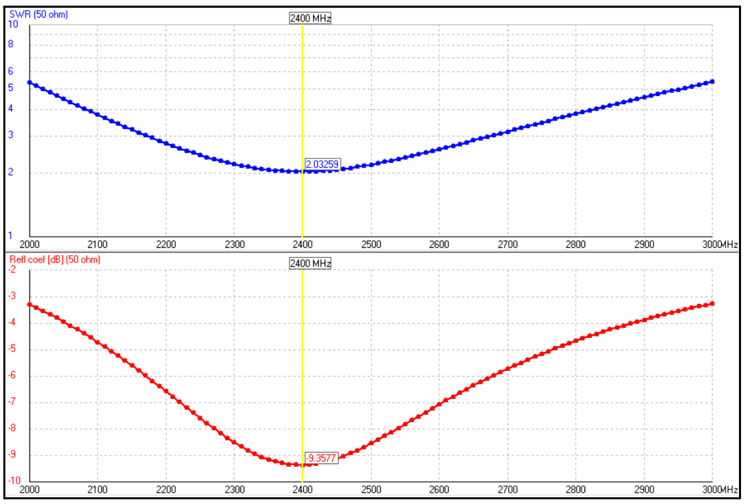


Figure 24: PIFA antenna second frequency scaling reflection coefficient graph.

Figures 12 and 13 present a worse antenna performance than those in the first iteration. The reflection coefficient only achieves a value of -9.36dB, below the industry standards for most applications, and the impedances are not particularly well matched either with 25.26Ω for the real part and -8.4Ω for the complex part. The results are both bad and expected, as we have reduced the antenna size by a huge percentage. However, we have previously seen awesome improvements over the frequency-scaled antenna during optimisation, so we hope the same thing will happen now. But before, we take a look at the radiation pattern.

1

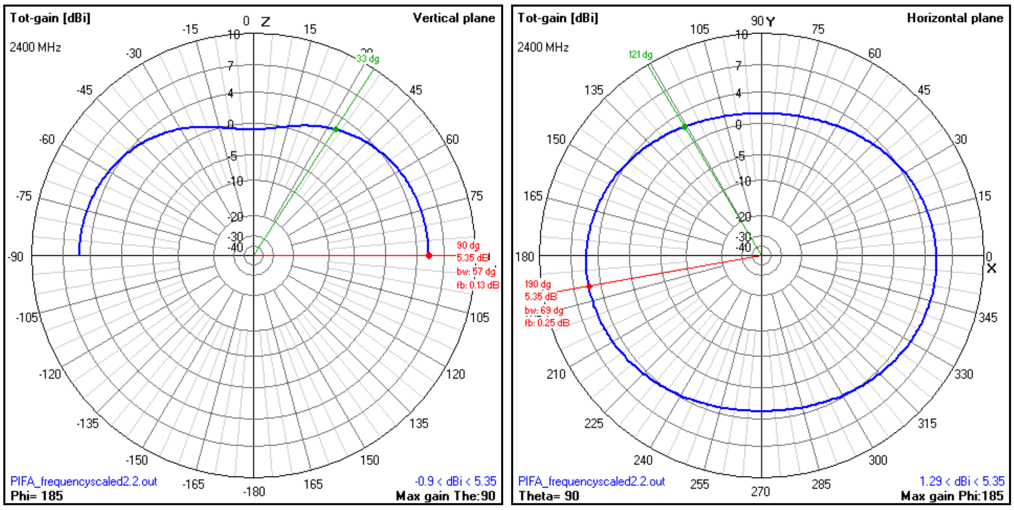


Figure 25: 2D Radiation diagrams - PIFA with second frequency scaling.

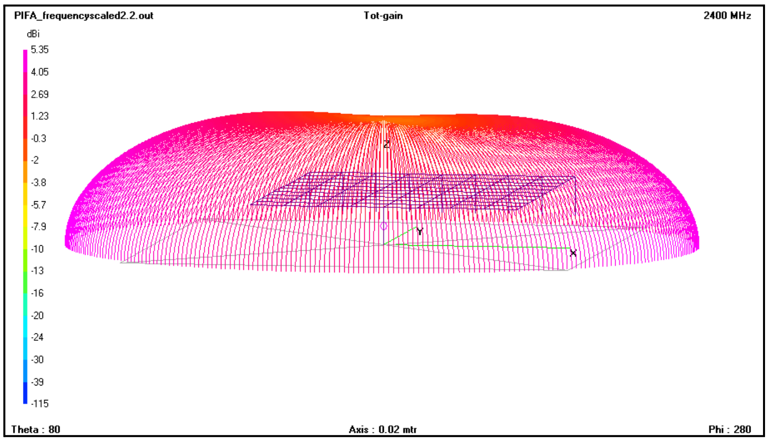


Figure 26: 3D Radiation diagram - PIFA with second frequency scaling.

The results are mostly what we would expect at this point. The changes are not that noticeable, with the biggest and most surprising one being the gain of 5.35dB, which is an improvement over the original antenna. The beamwidth has been reduced, however in an omnidirectional antenna beamwidth is not as relevant of a parameter, as it radiates very similarly in all of its non-zero directions, which is why we do not specify the numbers for it.

Then, we carried out optimisation, just as in the first iteration, optimising the metrics of SWR and Gain on the DW, DL and H variables, which yielded the following results:

Symbols

Nr	Symbols and equations
1	DW=3.909e-3
2	DL=3.156e-3
3	NL=10
4	NW=8
5	NS=8
6	L=DL*NL
7	W=DW*NW
8	H=8.871e-3
9	F=0
10	R=0.0005

Figure 27: PIFA antenna second optimisation NEC file.

Just as in the first iteration, comparing the sizes to those of the frequency-scaled antenna, we can see that DW and H have grown while DL has decreased. The implications of these changes in the performance of the antenna are dissected below:

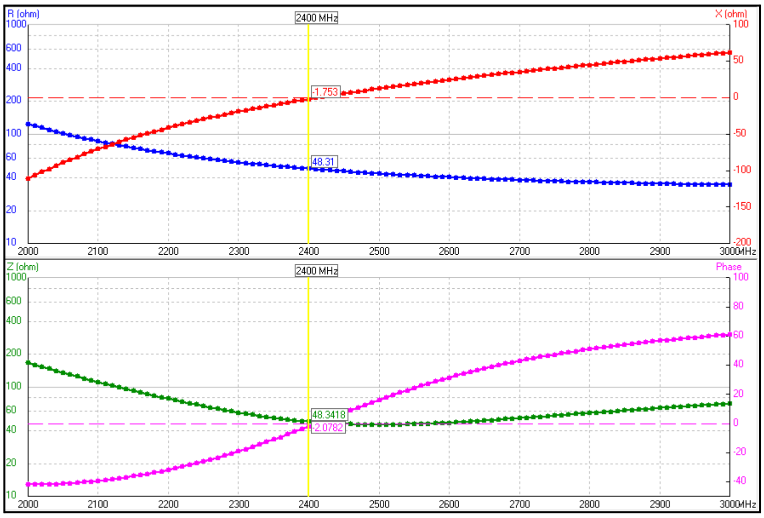


Figure 28: PIFA antenna second optimisation impedance graph.

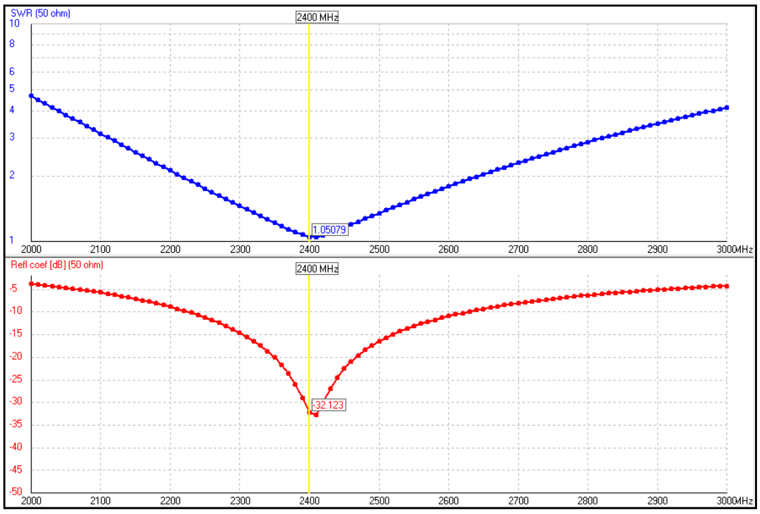


Figure 29: PIFA antenna second optimisation reflection coefficient graph.

Just as in the first iteration, the adaptation values after the optimisation shows great performance and improvement over the frequency-scaled version of our antenna in terms of impedance matching and coefficient of reflection. The impedances are very well matched at 2.4GHz with values of 48.31Ω for the real part and -1.7Ω for the complex part. Moreover, the coefficient reflection achieves a value of -32.12dB. Once again, these results are exceptional, well above the industry standard but, comprehensibly, slightly worse than the first iteration, due to the significant size reduction. Before extracting general conclusions on this second iteration we look at the Gain and radiation pattern:

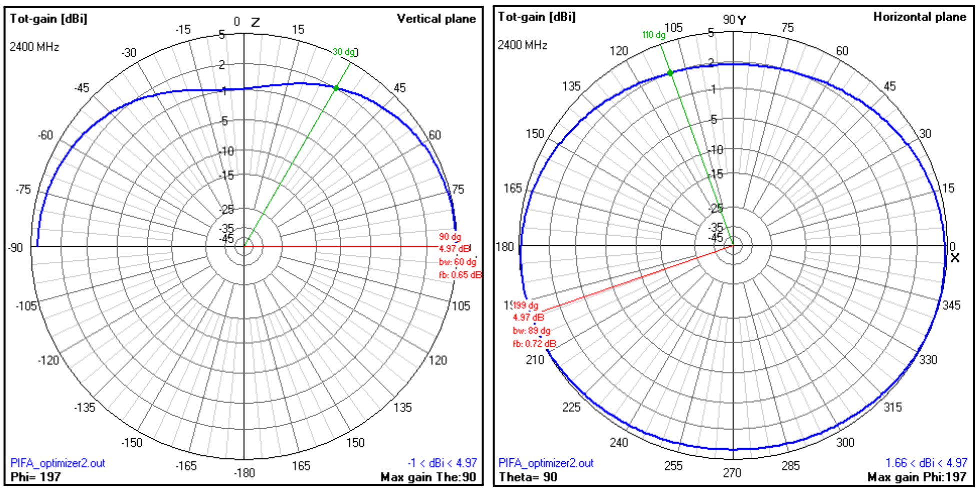


Figure 30: 2D Radiation diagrams - PIFA with second optimisation.

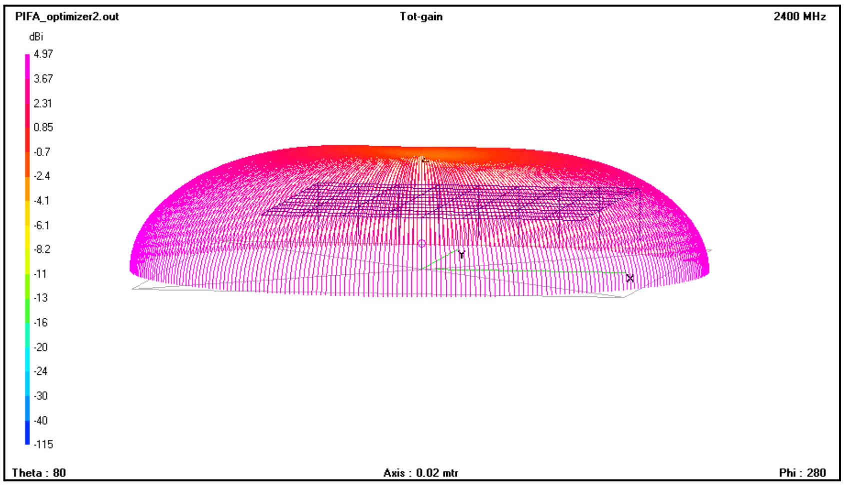


Figure 31: 3D Radiation diagram - PIFA with second optimisation.

Figures 30 and 31 present a consistent result compared to all the previous radiation patterns we have seen up to now, but they do have some interesting points to consider.

The main one is the maximum gain, which has decreased from 5.35dB in the frequency-scaled version to 4.97dB in the optimised version.

However, having seen a substantial improvement in the adaptation to 2.4GHz of the antenna, it was something to expect that it has a price to pay in radiation power.

On the other hand, a trend that has been going on less noticeably but that becomes clear now in the radiation patterns is that the centre of the half sphere in Figure 31 has been consistently losing power as we reduce the dimensions of the radiating unit of the antenna. This can also be observed comparing the vertical plane of Figures 8 and 30 as the 0º point of the default values antenna was at 2dB and it has now dropped to 0dB. This is also highlighted in the 3D view, by focusing on the top of the half-sphere in Figure 31 and realising it has become more yellow and less purple.

We considered these two details as a warning that we were reaching the point in antenna size where its further reduction would start affecting more considerably its performance. We were satisfied with the results of this second iteration, as they allowed us to make our antenna very small already, but we realised that the nice thing about simulation is that it allows you to iterate over many versions of the work very fast. For that reason, we decided to go one step further and carry out one last iteration of the process to reduce size and optimise the antenna.

2.3.3. Third iteration

This third iteration was our last attempt at reducing further the size of the antenna while maintaining its good performance in terms of gain and reflection coefficient. The difference between the second and third iterations is that, in this case, we did not start from scratch with the original values, but instead, decided to iterate over the previous version of the antenna. What that means is that we performed the manual cut over the last optimisation of the antenna, as, opposite to the first iteration, we were sure the second antenna did not carry any errors.

To perform the manual cut we reduced NW and NS to 6. This is a 70% decrease over the original antenna size. The variables then look as:

Symbols

Nr	Symbols and equations
1	DW=3.909e-3
2	DL=3.156e-3
3	NL=10
4	NW=6
5	NS=6
6	L=DL*NL
7	W=DW*NW
8	H=8.871e-3
9	F=0
10	R=0.0005

Figure 32: PIFA antenna third manual cut NEC file.

One can easily see that this last iteration only reduces NW and NS from 8 to 6. We decided to be less radical with this last manual reduction because the results of the second optimisation were great, but already presented some signs of the effect on performance that the considerable size reduction had caused.

Having clarified this aspect, we move on to the second step in frequency scaling. To do so, as we have already explained twice it was enough to realise that the manual cut antenna was better adapted to work at 2.57GHz, which implies a scaling factor of 2.565/2.4=1.069. After scaling it once, and realising that we could scale twice for optimal adaptation using a scale factor of 2.39/2.4=0.996, the results were the following:

Symbols

Nr	Symbols and equations
1	DW=4.161e-3
2	DL=3.359e-3
3	NL=10
4	NW=6
5	NS=6
6	L=DL*NL
7	W=DW*NW
8	H=9.443e-3
9	F=0
10	R=0.0005

Figure 33: PIFA antenna third frequency scaling NEC file.

It is a good moment to see that DW, DL and H are all bigger than the original values. Understandably, though, maintaining a good performance sort of "pushes" the antenna to become bigger, and the variables that can be fine-tuned are the ones "pushed" in that direction. The adaptation results of the frequency-scaled antenna are the following:

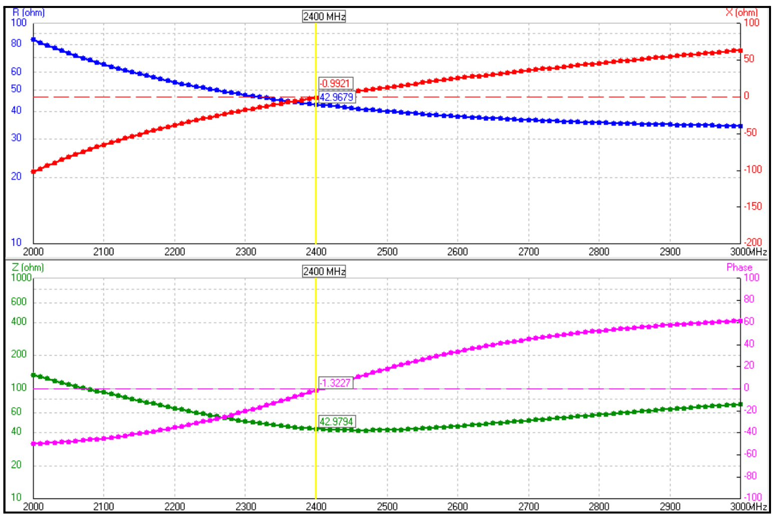


Figure 34: PIFA antenna third frequency scaling impedance graph.

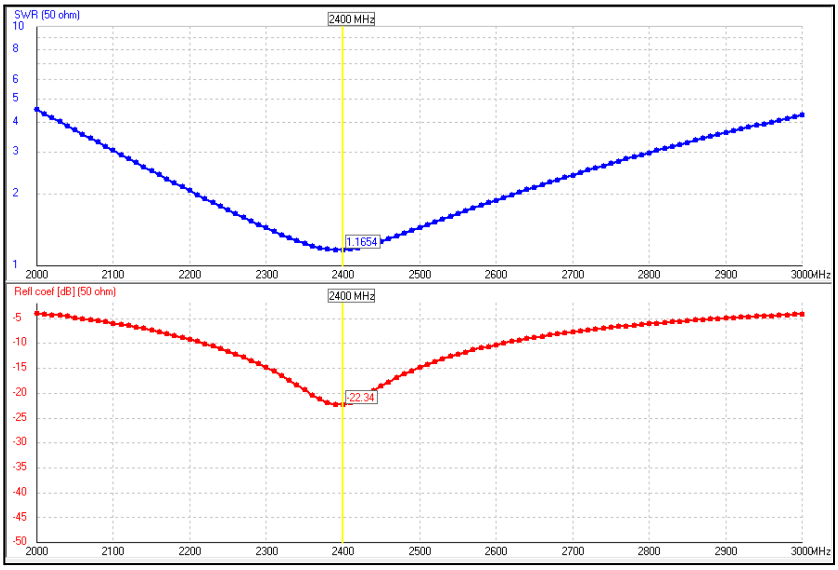


Figure 35: PIFA antenna third frequency scaling reflection coefficient graph.

Surprisingly, this antenna is much better adapted than the two previous frequency-scaled versions. It is easy to see that the responsible for this might be the optimiser, which adapted very well to the second frequency scaled antenna and gave us a great antenna to start with in this third iteration.

The reflection coefficient achieves a value of -22.34dB, aligned with the industry standards for most applications, and the impedances are quite well matched with 42.87Ω for the real part and -0.99Ω for the complex part.

The results of the frequency scaling up to now are very promising, as previous iterations relied strongly on the optimisation for good performance, but in this, we already can observe a good performer. Before carrying out the optimisation we look at the radiation patterns.

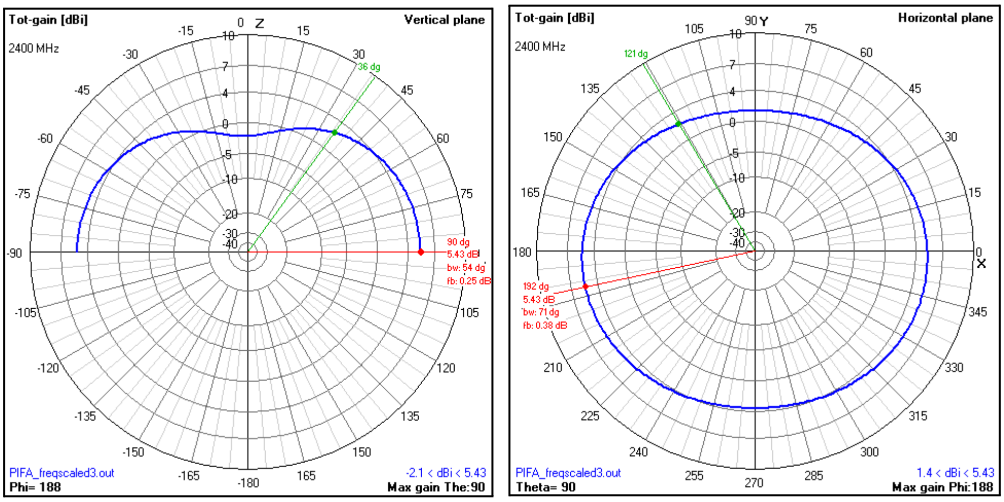


Figure 36: 2D Radiation diagrams - PIFA with third frequency scaling.

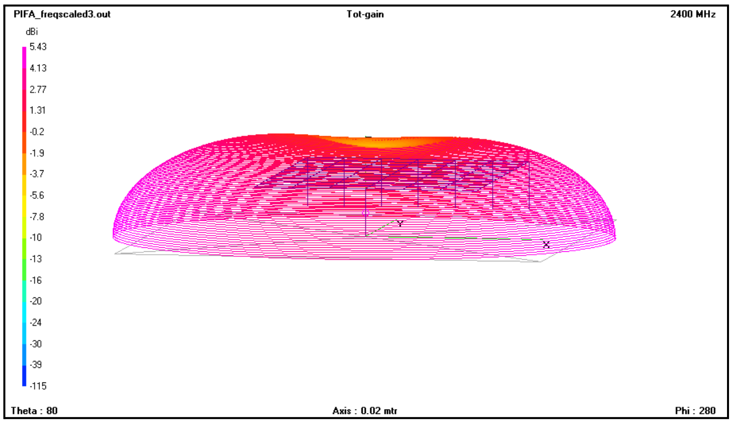


Figure 37: 3D Radiation diagram - PIFA with third frequency scaling.

The radiation patterns of Figures 36 and 37 present the same similarities to those of Figures 30 and 31. Performance is mostly consistent, even though the maximum gain is now at 5.43dB and Figure 37 shows an even deeper dent on the upper point of the radiation diagram, characterised with a yellower tone. These changes are consistent with what we observed in the previous iteration, so we move on to optimisation to evaluate the results of this last iteration.

As always, we optimise Gain and SWR for the variables DW, DL, and H. The results are the following:

Symbols

Nr	Symbols and equations
1	DW=4.24e-3
2	DL=3.227e-3
3	NL=10
4	NW=6
5	NS=6
6	L=DL*NL
7	W=DW*NW
8	H=9.985e-3
9	F=0
10	R=0.0005

Figure 38: PIFA antenna third optimisation NEC file.

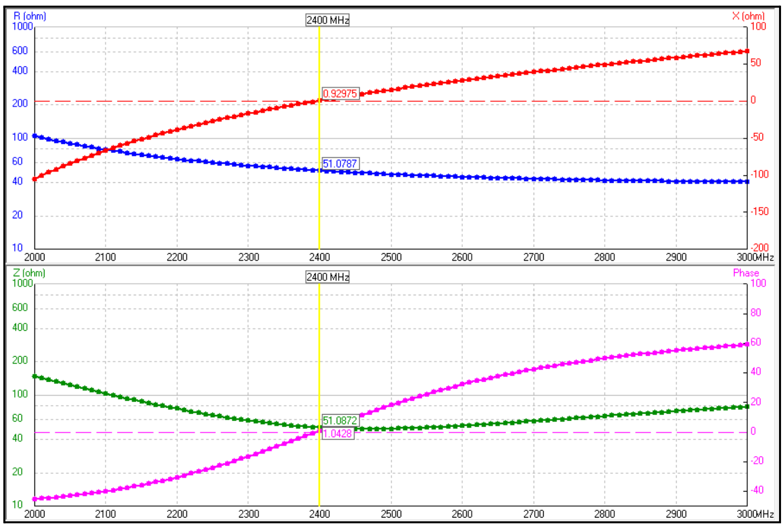


Figure 39: PIFA antenna third optimisation impedance graph.

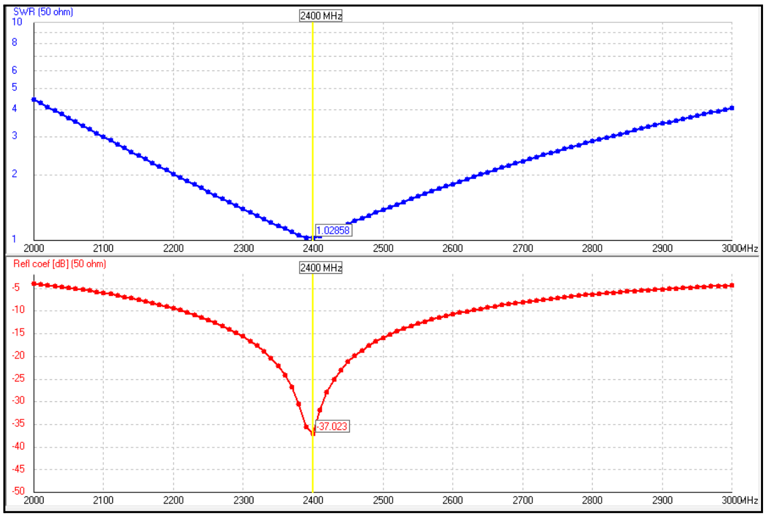


Figure 40: PIFA antenna third optimisation reflection coefficient graph.

Consistently with all the other optimisations, Figure 38 shows that DW and H have grown while DL has decreased. As for the adaptation of this last antenna, the results are extremely satisfactory, with the coefficient of reflection achieving a value of -37.02dB, clearly over the industry standards for all applications, and the impedances are very well matched with 51.07Ω for the real part and 0.93Ω for the complex part.

Before making the final judgement on this last optimisation, we also consider the radiation diagrams below:

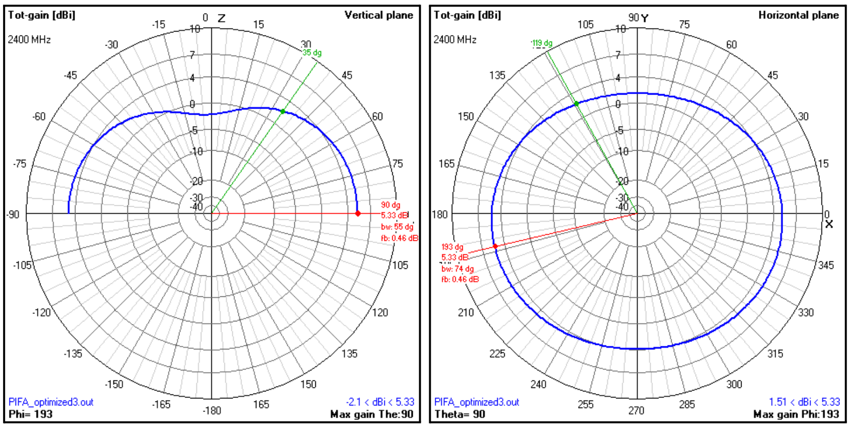


Figure 41: 2D Radiation diagrams - PIFA with third optimisation.

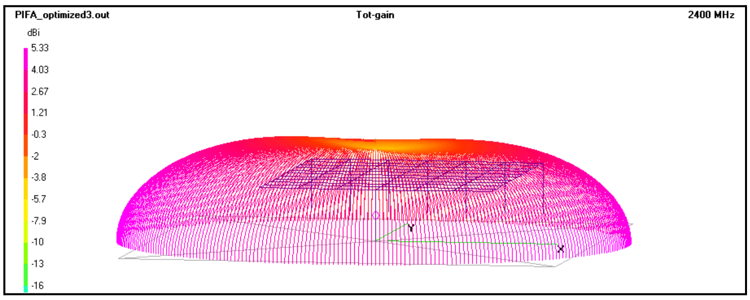


Figure 42: 3D Radiation diagram - PIFA with third optimisation.

Figures 41 and 42 present excellent results concerning radiation patterns. As always, most parameters have stayed the same, with the general shape of the pattern being preserved throughout the different iterations. The final maximum gain is 5.33dB, considerably better even than the maximum gain of the original antenna at 4.94dB. The rest of the parameters have stayed mostly the same, with the beamwidth being not-so-relevant for the reasons already mentioned and the indentation becoming even more noticeable thanks to the vertical diagram in Figure 41 and the yellow tones in Figure 42.

This simulation section concludes that the third iteration is clearly the best. We have insisted throughout the process that our main goal was to reduce the size of the antenna as much as possible while still retaining good performance characteristics. From our point of view, we did that very successfully, even improving in reflection coefficient and maximum gain over the values of the original antenna.

The process we designed to carry out this progressive reduction in size, advised by our professor, truly helped us understand how a PIFA antenna behaves and how to fine-tune its parameters to achieve the desired results. Also, what the implications of reducing the antenna size are, particularly in the radiation pattern domain.

That being said, we acknowledge that the numbers yielded by the last optimisation, which is the one we aim to recreate in the following sections, are subject to a set of conditions previously specified which do not translate directly into real life. Therefore, even though we expect good performance, we know we will not be measuring reflection coefficients of around -40dB with a 70% size decrease, despite how good our precision during the fabrication process is, but more on that in the following sections.

3. Fabrication process of the antenna

3.1. Understanding our needs

The fabrication process for the PIFA antenna was relatively straightforward compared to the Yagi-Uda antenna from the previous lab. The primary goal was to create a compact antenna with a small form factor while maintaining good performance. The PIFA antenna consists of a radiating patch, a ground plane, a shorting plate and a feed point, all of which needed to be carefully designed and assembled.

3.2. Materials and Tools

To fabricate the PIFA antenna, we used the following materials and tools:

- **Copper sheet:** For the radiating patch and ground plane.
- **SMA connector:** To feed the signal to the antenna.
- **Precision cutting and measuring tools:** To ensure accurate dimensions for the radiating patch and ground plane.
- **Soldering iron and solder:** For connecting the SMA connector and other components to the antenna.

3.3. Fabrication Steps

1. **Design Transfer:** The dimensions obtained from the simulation (after the third iteration) were transferred to the copper sheet. The radiating patch and the ground plane were cut to the specified size.

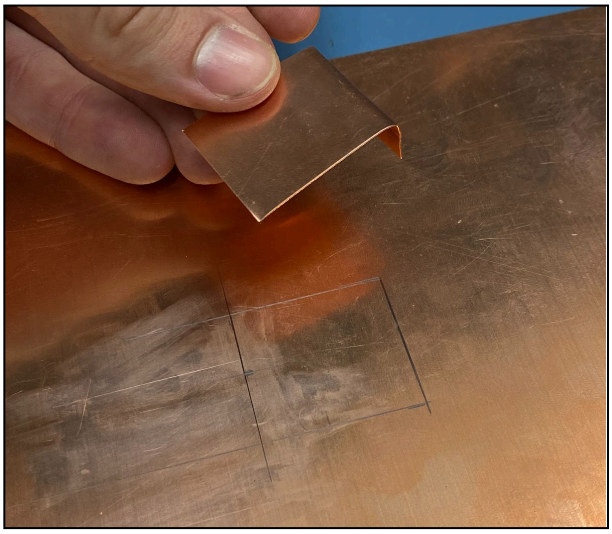


Figure 43: Cutting and folding the copper sheet

This step was one of the most critical yet challenging parts of the fabrication process. Due to the lack of highly precise tools and materials, we anticipated that human error would play a significant role in the outcome. To minimise inaccuracies, we took extra care during the cutting process, ensuring that the dimensions were as close as possible to the simulated design.

The radiating patch required an additional step: one side had to be folded at a 90° angle to create the shorting plate, which is the connection point that would later be soldered to the ground plane. Achieving a precise fold was essential to maintain the antenna's structure and performance.

2. **Feed point hole:** After cutting and folding the radiating patch, the next step was to drill a hole in the ground plane for the feed point. This hole would later serve as the connection point for the SMA connector, the monopole, and the radiating patch. The positioning of the hole was critical, as it needed to align perfectly with the feed point on the radiating patch to ensure proper signal transmission and impedance matching. We tried to ensure the hole was clean and free of burrs, as any irregularities could affect the connection quality.

3. **Soldering step:** The SMA connector was soldered to the feed point using solder, allowing the antenna to be connected to measurement equipment.

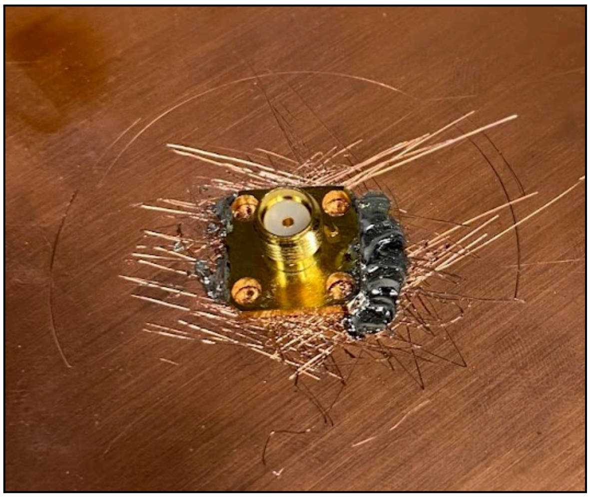


Figure 44: SMA connector soldering

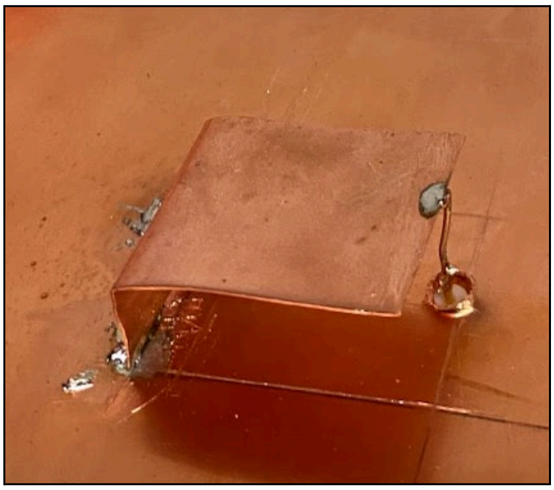


Figure 45: Shorting plate and monopole soldering

Soldering was another challenging aspect of the fabrication process. The materials used for both the antenna's structure and the soldering process proved difficult to work with, primarily due to the smooth surface of the copper sheet. The smoothness of the copper made it difficult for the solder to adhere properly, which could lead to weak or unreliable connections.

To address this issue, we scratched all the surfaces that needed to be soldered. This created a rougher texture, which improved the adhesion of the solder and facilitated a stronger bond.

4. Measurements of the antenna

4.1. Measuring the PIFA with VNAJ

After the fabrication process, the next step was to measure the performance of the PIFA antenna using a Vector Network Analyzer (VNA). The primary goal was to evaluate the antenna's reflection coefficient and impedance matching at the target frequency of 2.4 GHz.

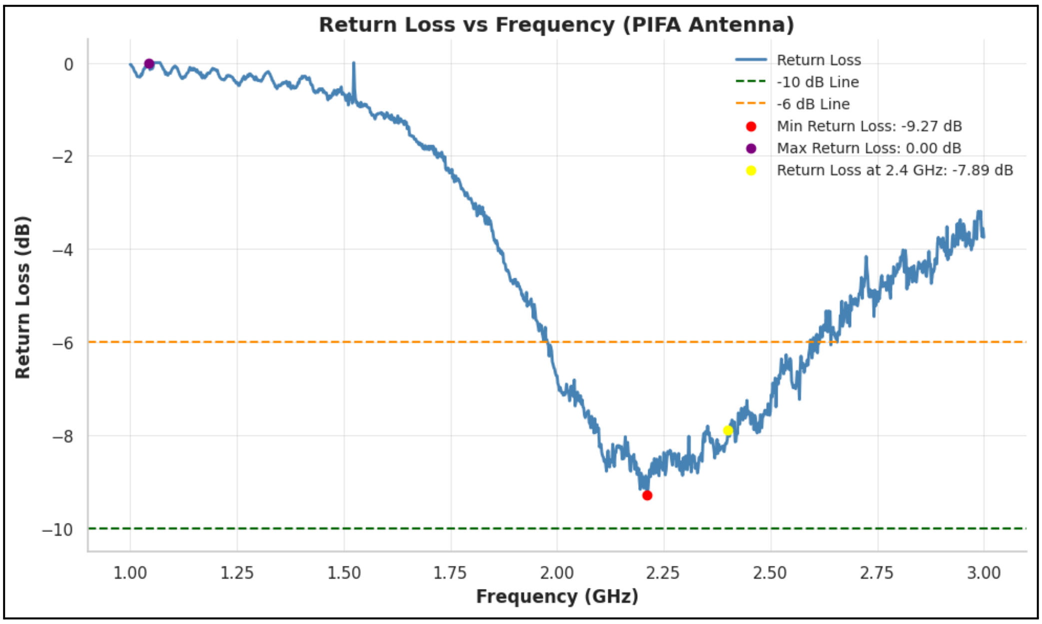


Figure 46: PIFA antenna Vnaj measurements

The VNA measurements revealed the following results:

- Return Loss: The antenna achieved a return loss of -9.27 dB at 2.2 GHz, which is below the industry standard of -10 dB for most applications. This indicated that the antenna was not perfectly matched to the 50Ω impedance, likely due to fabrication imperfections or deviations from the simulated design.

- Frequency Response: The return loss graph showed that the antenna resonated close to the desired frequency, but the performance was not optimal. The maximum return loss was 0.00 dB, and the return loss at 2.4 GHz was -7.89 dB.

The measured results showed significant deviations from the simulated design, particularly in the reflection coefficient and impedance matching. These discrepancies are likely due to fabrication tolerances, such as slight errors in dimensions, soldering quality, or material properties. Despite these issues, the antenna still resonates close to the target frequency and demonstrated usable performance within the -6 dB bandwidth.

4.2. Carrying out a Wi-Fi test with our PIFA

The last part of our measurements involved testing the performance of the PIFA antenna in a real-world environment. To do this, we used a router to emit a Wi-Fi signal, and a computer program to calculate the signal strength received.

We began by connecting a regular monopole antenna to the router and measured the Wi-Fi signal strength using the program. After obtaining baseline measurements, we disconnected the monopole antenna, waited for some seconds and replaced it with our PIFA antenna using the SMA flange connector soldered to the antenna.

As we explained in the last lab, a monopole antenna is a simple, omnidirectional antenna consisting of a single conductor mounted over a ground plane, commonly used in mobile communications and Wi-Fi applications. It radiates power equally in all horizontal directions, making it useful for broad coverage but resulting in a relatively low gain (typically around 2–5 dBi). During the last laboratory, we compared this antenna against a Yagi-Uda, which is significantly more directional and provides a significantly higher gain (8–15 dBi). However, in this case, we will compare it against a PIFA antenna, which is characterised by performance figures very similar to those of a monopole antenna, being omnidirectional and also having a gain of around 2-5 dBi.

Therefore, the results of this comparison are much more directly comparable and can provide more insights. They are presented below:

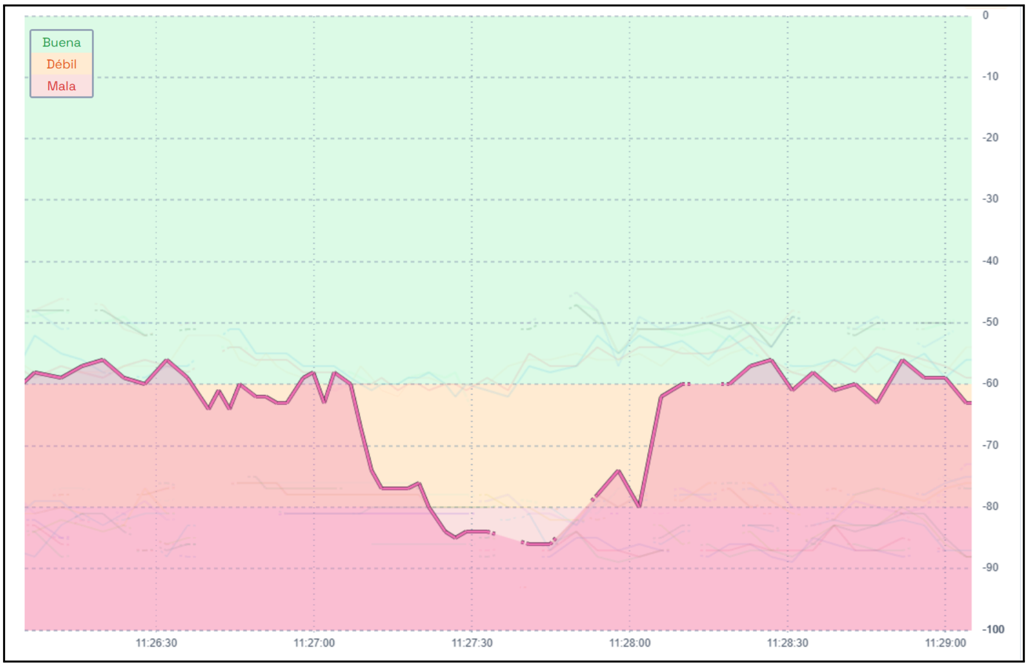


Figure 47: Result of the Wi-Fi test measurements

The graph illustrates the received Wi-Fi signal strength under three different conditions:

1. First section (first third of the graph): The Wi-Fi signal was measured while using a monopole antenna connected to the router.
2. Middle section (second third of the graph): The measurement was taken with no antenna connected to the router.
3. Final section (last third of the graph): The signal strength was recorded when the PIFA antenna was connected to the router.

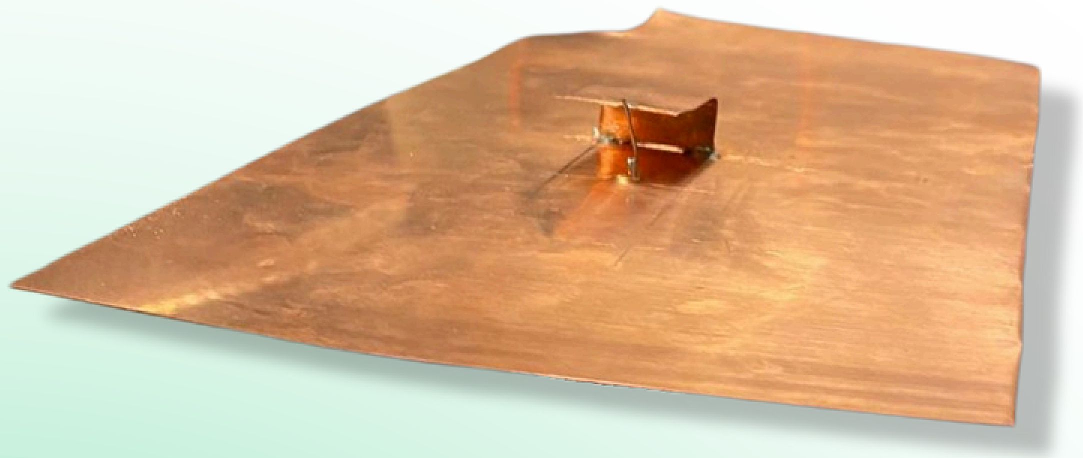
The results align with our expectations. As we have previously introduced, the monopole antenna possesses very similar properties to those of the PIFA antenna, so it was expected to see similar performance from both of them.

Nonetheless, given the previous measurements with the VNAJ software and their very big difference with the simulation, we considered the possibility for the radiation pattern to also have a similar difference in measured performance. We were pleasantly surprised when we realised that this was not the case and it performed essentially exactly like the monopole antenna, and based on the power usually radiated by monopole antennas, it also performs quite similarly to the simulation as well (2-5 dBi).

On the other hand, if one considers the rest of the report, coming across this is not that exceptional, since, throughout the long process of simulation and optimisation of the antenna during its size reduction, we saw that the radiation pattern was something which remained almost unaltered. Therefore, seeing that the real-life limitations, such as cut precision and environmental factors do not have as big of an effect on radiated power as they do on the reflection coefficients is something that aligns well with the simulation and the rest of our work.

Overall, we are satisfied with the performance of our antenna, given its comparability in this test to a standard-produced monopole antenna. Despite the limitations we encountered in the previous section for the reflection coefficient, it is great to see that it is not that hard to produce a decent WiFi antenna with very simple components and no advanced machinery whatsoever.

5. Datasheet of the antenna



2.4GHz PIFA Antenna Datasheet

A compact, high-efficiency PIFA antenna designed for 2.4GHz WiFi applications, offering low-profile integration and optimized radiation for reliable wireless performance.

Presented by

José Pérez Clar
Ricardo Pereira González

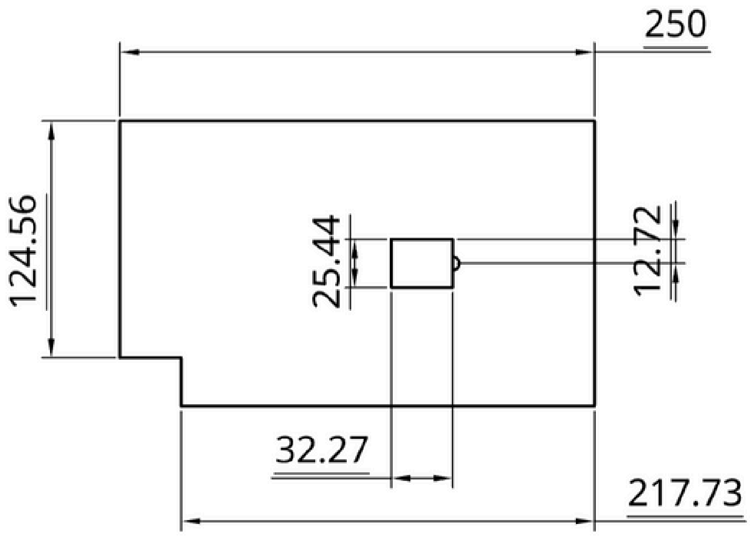
Presented to

Radiocommunication Systems - 2025

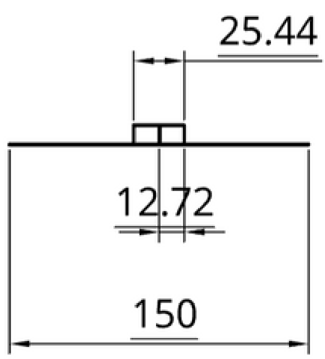
P&P ANTENNAS

Dimensions of the antenna

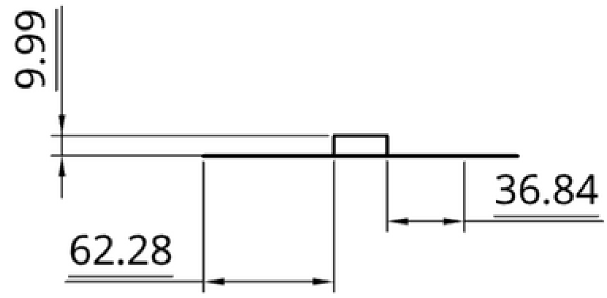
TOP



RIGHT



LEFT



Important Considerations

- All the distances are stated in millimetres (mm).
- The drawings above are not scaled.
- The connector hole has a diameter of 7 mm.
- The ground plane is approximated to 250x150mm.
- All values concerning the feed point and radiating patch are exact.

Behavioural Details

Frequency Band: 1.9 GHz to 2.65 GHz (at -6 dB)

Adaptation Values and Graphs

Return Loss:

- Minimum Return Loss: -9.27 dB
- Maximum Return Loss: 0.00 dB
- Return Loss at 2.4GHz: -7.89 dB

Impedance at 2.4GHz:

- Real Part (R): 51.0787 Ω
- Imaginary Part (X): 0.9298 Ω

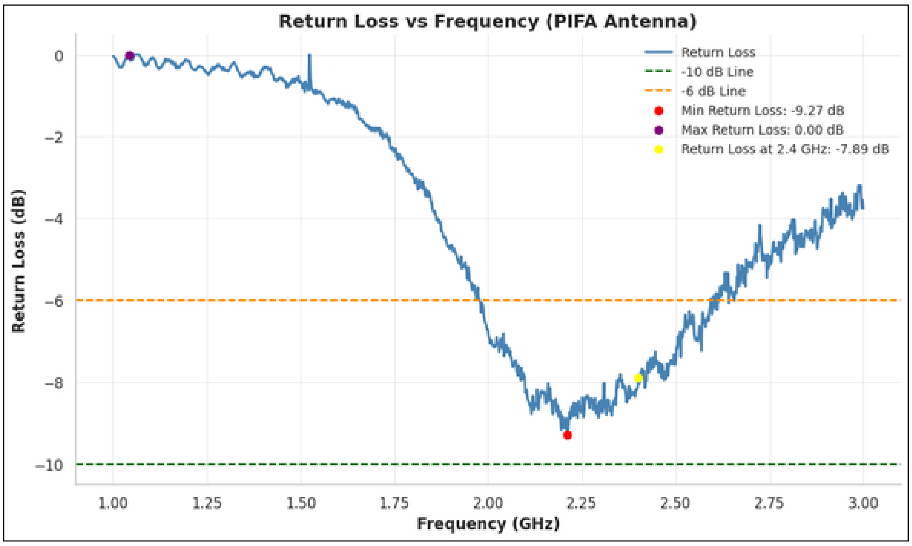


Figure 1: Return Loss graph

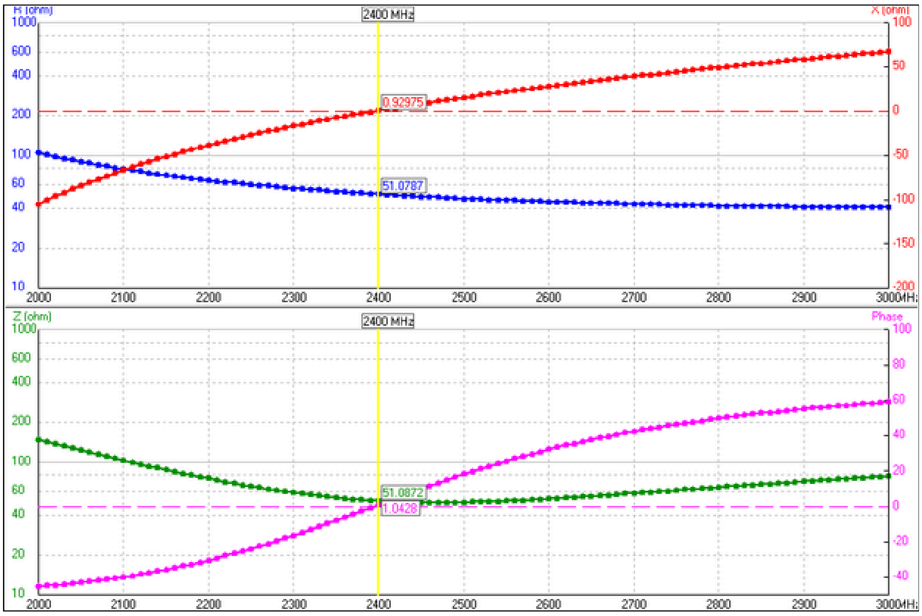


Figure 2: Impedances graph

Behavioural Details

Radiation Data at 2.4GHz

Max Gain: 5.33 dBi

Front-to-Back Ratio (F/B): 0.46 dB

Beamwidth:

- Vertical Plane (Elevation): 55°
- Horizontal Plane (Azimuth): 74°

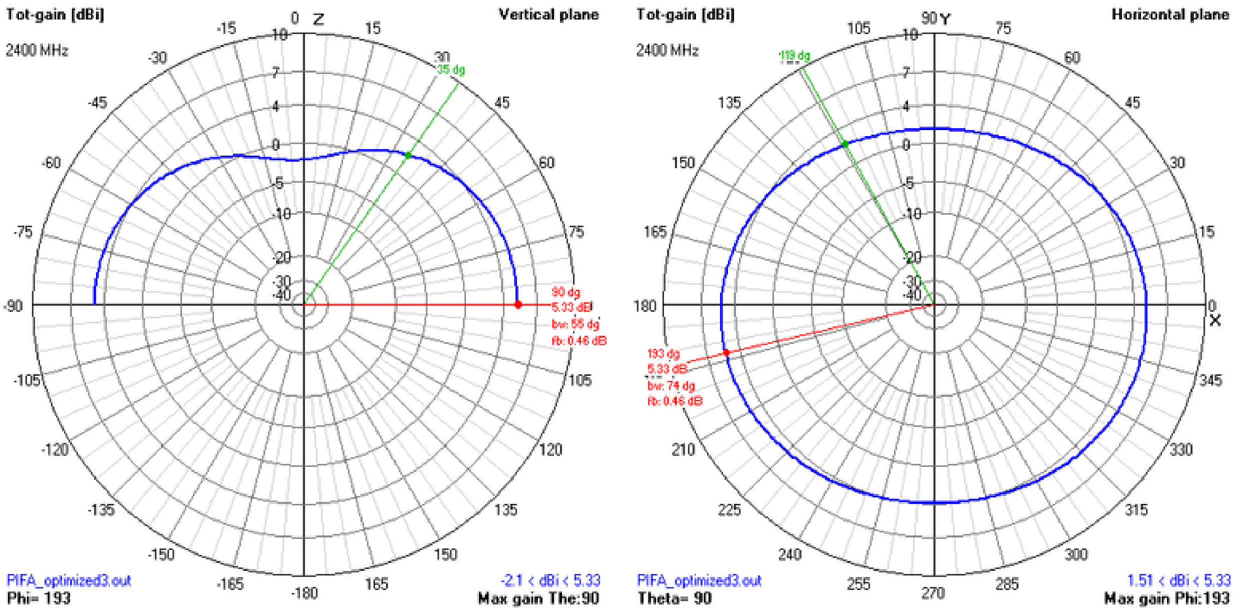


Figure 3: 2D Radiation diagrams

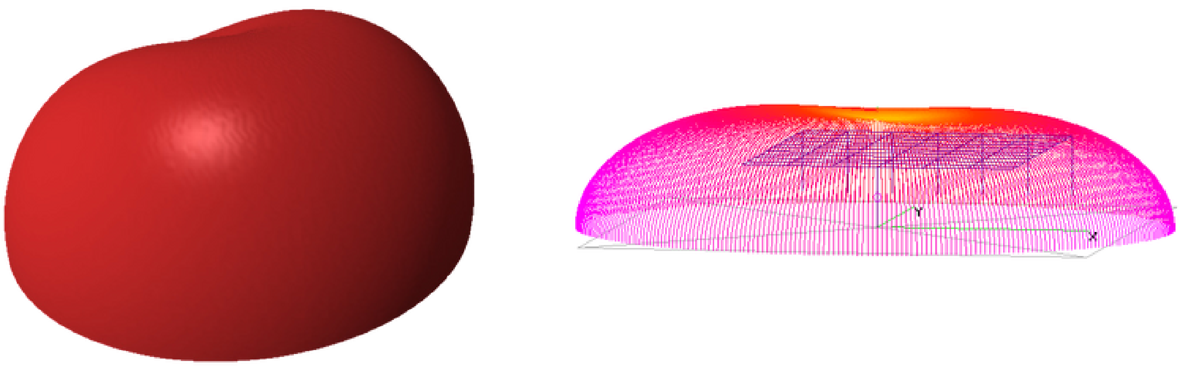


Figure 4: 3D Radiation diagrams

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Questions? Contact us.

6. Conclusion

This laboratory has allowed us to carry out a detailed exploration of the Planar Inverted-F Antenna (PIFA), covering its theoretical foundations, practical design, fabrication, and measurement. Through this process, we have gained valuable skills in antenna design and understood once again the challenges of translating simulations into real-world applications.

We began by looking into the fundamental principles of the antenna, contrasting wire antennas like the Yagi-Uda with patch antennas like the PIFA. The PIFA was chosen for its compact form factor and omnidirectional radiation pattern, making it suitable for modern wireless communication devices.

Using the 4NEC2 simulation software, we modelled and iteratively optimised the PIFA design to operate efficiently at 2.4 GHz. The process involved size reduction while maintaining good radiation characteristics. However, as observed in previous projects, simulations do not perfectly reflect real-world conditions, with fabrication tolerances and material imperfections impacting final performance.

The fabrication phase reminded us of the needed precision in cutting, folding, and soldering the copper components, as even minor deviations affected the antenna's response. Despite these challenges, the Wi-Fi test demonstrated that the PIFA performed similarly to a standard monopole antenna, confirming its practical utility.

In conclusion, this project deepened even further our knowledge of antenna engineering and allowed us once again to construct, from scratch, what at first seemed like very complex communication device components.

7. References

All the sources for this essay's content are cited below.

- Antenna Theory: Analysis and Design - Constantine A. Balanis
- Garg, R., Bhartia, P., Bahl, I., & Ittipiboon, A., Microstrip Antenna Design Handbook, Artech House, 2001
- pifa. (n.d.). https://docs.exponenta.ru/antenna/ref/pifa.html
- E, Z., Fu, D., Yu, Z., Hu, Y., & Nie, Y. (2021). Transformer condition monitoring technology based on surface acoustic wave passive wireless sensing antenna. E3S Web of Conferences, 257, 01085. https://doi.org/10.1051/e3sconf/202125701085
- Wikipedia contributors. (2024, August 31). Inverted-F antenna. Wikipedia. https://en.wikipedia.org/wiki/Inverted-F_antenna
- Wikipedia contributors. (2024, December 21). Omnidirectional antenna. Wikipedia. https://en.wikipedia.org/wiki/Omnidirectional_antenna
- Figure 1. The basic structure of the PIFA antenna [9], [23]. (n.d.). ResearchGate. https://www.researchgate.net/figure/The-basic-structure-of-the-PIFA-antenna-9-23_fig1_359555525

LLMs were not used to create original content, introduce assumptions or generate independent ideas. Its role was limited to refining the wording of existing text for clarity and cohesion.